

Fig. 12: Thermistor attached to heat-bed



Fig. 13: Extruder assembly

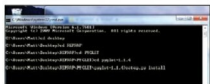


Fig. 14: Installing Pyglet

various operations. The overall system is powered by 12V, 400W ATX power supply (Fig. 8).

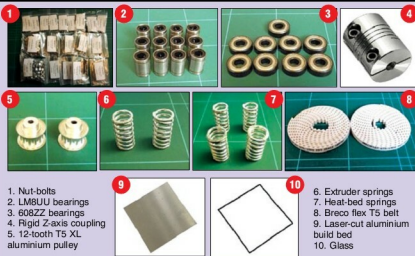
Sanguinololu board. Sanguinololu board (shown in Fig. 9) is a low-cost, all-in-one electronics solution for Reprap and other CNC devices. It features an onboard Sanguino clone using the ATMEGA644P microcontroller, though an ATMEGA1284 can be easily substituted. The board is developer-friendly with expansion pins supporting PC, SPI, UART and ADC functions. Sanguinololu has a very flexible input power supply that ranges from 7V to 30V.

Pololu board. The stepper motors are powered by Pololu boards that are mounted over Sanguinololu board. These are carrier boards or breakout boards for Allegro's A4988 DMOS microstepping driver with translator and over-current protection. The stepper motor driver lets you control a bipolar stepper motor at up to 2A output cur-

Electronic/electrical parts



Mechanical parts



1. Nut-bolts
2. LM8UU bearings
3. 608ZZ bearings
4. Rigid Z-axis coupling
5. 12-tooth T5 XL aluminium pulley

Printed parts



rent per coil. Fig. 10 shows a Pololu board with heatsink.

Heat-bed. The heat-bed (Fig. 11) normally heats up to 110°C when powered through the dedicated connection on Sanguinololu board. The power sup-

ply should be able to deliver at least 300W and wires from the power supply to the Sanguinololu board should be capable of handling 20A or slightly more current.

Thermistor. A thermistor is at-